2.4 Stage 2: voxel-wise Bayesian tumor segmentation

From localization to segmentation. The two stages answer different questions. Stage 1 is *localization*. A tumor agent can tell that a region looks abnormal but not where it lies anatomically, nor whether that location makes sense for the cancer; this needs the anatomy agents and the knowledge base. Stage 1 therefore reasons globally, over relations between tumor units and organs, and outputs for every lesion L a coarse decision (accepted, uncertain or rejected), a set of anchors and candidate extents, and a host credible set $\mathcal{S}_L$ with probabilities $P(h \mid L)$. Stage 2 is *segmentation*. It works locally, voxel by voxel, on every accepted and uncertain lesion: where exactly the tumor is, which voxels the agents included wrongly or missed, and how the tumor is composed. The agents recognise tumor as a whole; they do not delineate its internal structure.

Between the stages the role of the host changes. In stage 1 the host is unknown and inferred (Eq. normal model, Section 2.1); in stage 2 it is *known with uncertainty*: $P(h \mid L)$, $h \in \mathcal{S}_L$, renormalised over the credible set, is a fixed prior, and the anatomy stays cut. Stage 1 does not fix the tumor at voxel level. Its extents are proposals drawn by the agents, and its decision concerns the lesion as a unit. The tumor mask is therefore a latent variable of stage 2, with a prior learned from expert data, and the stage-1 extent serves only as initialisation and as one input to that prior.

Region of interest. The model is defined on a region combining a local neighbourhood of the lesion with all of its credible hosts, because the host is uncertain and a tumor may extend into or arise from any of them:

$$R_L = \Big(\bigcup\nolimits_{a \in L} a\Big) \oplus r \;\cup\; \bigcup\nolimits_{h \in \mathcal{S}_L} \tilde{\Theta}_h, \tag{1}$$

restricted to the body. $\tilde{\Theta}_h$ is the territory of h, joined with its knowledge-base region when h is unsegmented or its mask is below the knowledge-base volume range (an organ largely replaced by tumor). The radius r is the 95th percentile, over expert tumors, of the distance from a simulated partial core to the farthest tumor voxel. If the posterior tumor reaches the boundary of R_L inside the body, R_L is enlarged by r around it and inference is repeated, so that a tumor much larger than its anchor is not cut by the region.

Generative model. Let y_v be the image features of voxel $v \in R_L$: the intensity, smoothed in the acquisition plane, and local means and standard deviations at two scales. Each voxel carries a latent label $Z_v \in \{0, 1, \ldots, K\}$: 0 is background, $1, \ldots, K$ are tumor components (for example enhancing, necrotic or cystic tissue). The tumor indicator is $t_v = \mathbf{1}[Z_v \geq 1]$, and the tumor mask is $\{v : t_v = 1\}$.

Spatial tumor prior. Tumors are not spatially homogeneous: they arise in particular organs, extend beyond the agents' outlines by characteristic distances, invade neighbours and respect some boundaries. A homogeneous label prior, as used for organ segmentation, does not express this. The prior probability that a voxel is tumor is

$$\pi_v = P(t_v = 1) = \sum_{h \in \mathcal{S}_L} P(h \mid L)\, \sigma\big(\eta^\top \psi_v(h)\big), \tag{2}$$

a mixture over the credible hosts. The features $\psi_v(h)$ are:

- the signed distance of v to the stage-1 extent $\hat{\xi}_L$;
- the stage-1 extent marginal $q_v = \sum_\xi w^0(\xi)\, \mathbf{1}[v \in \xi]$;
- whether v lies in the territory of h, in another credible host, or in a confidently segmented organ outside $\mathcal{S}_L$;
- the depth of v below the surface of that organ.

The weights η are fitted on expert data only: expert tumors with simulated agent outlines (partial cores, dilated and leaking outlines, outlines merged with the organ, lost hosts) and expert organ masks. Eq. (2) is therefore the frequency with which an expert tumor occupies a voxel in that position relative to an agent outline and to the anatomy. Unlike the extent prior of stage 1, it gives positive probability beyond the agents' outlines, which is what allows recall.

Edge-aware spatial coupling. Neighbouring labels are coupled by a Potts term whose weights are latent, as in BayeSeg [Gao et al., BayeSeg]:

$$p(Z \mid \omega) \;\propto\; \prod_v \pi_v^{t_v} (1 - \pi_v)^{1 - t_v} \prod_{v \sim w} \exp\big(-\omega_{vw}\, s_{vw}\, \mathbf{1}[t_v \neq t_w]\big), \qquad \omega_{vw} \sim \text{Gamma}(a_\omega, b_\omega), \tag{3}$$

where s_{vw} scales a neighbour pair by its physical distance, so thick slices couple less. The weight ω_{vw} is inferred: it becomes small where the image has an edge, so the tumor boundary can follow real edges rather than a fixed smoothness. The hyperparameters (a_ω, b_ω) are estimated on expert tumor masks, from the distribution of label changes along image gradients.

Clamp. The two-agent core of the lesion, eroded by one voxel, is fixed to tumor ($t_v = 1$). This conditions stage 2 on the lesion being a tumor. Whether it is one remains the stage-1 decision, and no prior on tumor existence is needed.

Appearance. Tumor appearance and background appearance are latent and specific to the scan. Tumors are heterogeneous and long-tailed (necrosis, calcification, haemorrhage), so each component has a Student-t likelihood:

$$p(y_v \mid Z_v = k) = \mathrm{St}\big(y_v \mid \mu_k, \Lambda_k, \nu\big),\ k = 1..K; \qquad p(y_v \mid Z_v = 0) = \sum_{c \in \mathcal{C}_L} \lambda_{v,c}\, \mathrm{St}\big(y_v \mid \mu_c, \Lambda_c, \nu\big). \tag{4}$$

The background is a mixture over the credible hosts' normal tissue and the surrounding anatomy, $c \in \mathcal{C}_L$. Its location weights $\lambda_{v,c}$ come from the organ territories, and the weight of host h is scaled by $P(h \mid L)$. The uncertainty about the host therefore carries into the voxel posterior instead of being resolved first. The means and precisions have conjugate Normal–Gamma priors. Their hyperparameters are estimated on expert data, per organ class and modality: the contrast of expert tumors relative to their organ's normal tissue and its spread, and the variability of normal tissue. The component weights have a Dirichlet prior. The number of tumor components $K \le K_{\max}$ is chosen per lesion by the evidence lower bound, with $K_{\max}$ the largest number supported on expert tumors. On the scan, the agents' voxels only initialise the appearance; it is inferred jointly with the mask.

Inference. The posterior $p(Z, \theta, \omega \mid y)$ is approximated by mean-field variational Bayes, $q(Z)\, q(\theta)\, q(\omega)$, with closed-form coordinate updates of the conjugate families [Gao et al., BayeSeg]. The label posterior is initialised from the stage-1 extent and the appearance from the voxels it implies. The outputs are:

- the *voxel tumor posterior* $p_v = q(t_v = 1)$, given that the lesion is a tumor;
- the *component posteriors* $q(Z_v = k)$, an intra-tumoral substructure map;
- the *refined tumor* $\hat{Z}_L$, the voxels with $p_v \ge \frac{1}{2}$ connected to the clamped core.

Comparing $\hat{Z}_L$ with the stage-1 extent gives the *trimmed* voxels $\hat{\xi}_L \setminus \hat{Z}_L$ (false positives of the agents' outline) and the *recalled* voxels $\hat{Z}_L \setminus \hat{\xi}_L$ (tumor the agents missed), each with the host territories it lies in. Lesions whose refined tumors meet are merged and the merge is reported. Recall comes from the image and the spatial prior only: the agents' discarded proposals (claims, crumbs and lobes that stage 1 did not keep) are never re-admitted.

Effect on the stage-1 evidence. Decisions are left unchanged. For every lesion, the stage-1 existence evidence (Section 2.3, Image grounding) is recomputed on $\hat{Z}_L$ under each host of $\mathcal{S}_L$, with the same expert tables and level α, and reported next to the stage-1 outcome, together with $\bar{p}$ over the stage-1 extent and over $\hat{Z}_L$. Whether the refined evidence should feed back into decisions is decided after this comparison.

Estimation and validation on expert data. Every prior and hyperparameter of stage 2 comes from expert annotations, never from agent outputs or the evaluated cohort:

- the spatial prior η;
- the edge hyperparameters (a_ω, b_ω);
- the appearance hyperparameters, per organ class and modality;
- the degrees of freedom ν;
- $K_{\max}$ and the radius r.

The data are R2Seg images with expert tumor masks, organ masks from TotalSegmentator and R2Seg, and simulated agent outlines. Validation holds out R2Seg collections in turn. It reports, per organ class and modality, voxel precision and recall, Dice, the over- and under-segmented fractions and the calibration of p_v, against the stage-1 extent, the anchor alone and the union of the agents' outlines. Stage 2 replaces the stage-1 extent only for the organ classes and modalities in which it improves the held-out Dice; elsewhere the stage-1 extent is kept, and this is reported.